

Chapter 16: Flat Clustering

Complete Study Notes — Teaching Session

Based on: Introduction to Information Retrieval

Manning, Raghavan & Schütze (Cambridge University Press)

Sections Covered:

- Chapter 16 Introduction — Clustering Overview
- 16.1 Clustering in Information Retrieval
- 16.2 Problem Statement
 - 16.2.1 Cardinality — The Number of Clusters
- 16.3 Evaluation of Clustering
 - 16.4.1 Cluster Cardinality in K-Means

Chapter 16: Flat Clustering — Introduction

What is Clustering?

Clustering algorithms group a set of documents into subsets called clusters, such that documents inside a cluster are as similar as possible to each other, while documents across different clusters are as dissimilar as possible. Think of it like sorting thousands of randomly scattered books — science books in one corner, history in another, fiction elsewhere.

Unsupervised Learning

	Supervised Learning (Classification)	Unsupervised Learning (Clustering)
Teacher/Expert?	Yes — human labels the data	No — no human labels
Goal	Replicate human-assigned categories	Discover natural groups in data
Example	This email IS spam (pre-labeled)	These emails LOOK similar — group them
Chapter	Chapter 13	Chapter 16 & 17

The Distance Measure

The most important input to any clustering algorithm is the **distance measure**. Smaller distance = more similar documents = likely same cluster. Larger distance = more different = likely different clusters. In document clustering, Euclidean distance and cosine similarity are most common. **Different distance measures give different clusterings!**

Flat vs Hierarchical Clustering

Type	Description	Analogy
Flat Clustering (Ch.16)	Simple flat set of groups; no relationship between clusters	Sorting books into piles on a table
Hierarchical (Ch.17)	Tree/hierarchy of clusters; clusters contain sub-clusters	Shelves → sections → categories

Hard vs Soft Clustering

	Hard Clustering	Soft Clustering
Assignment	Each doc belongs to exactly ONE cluster	Doc has fractional membership in multiple clusters
Membership values	0 or 1 only	Any non-negative value (probabilities)
Example algorithm	K-Means	EM (Expectation Maximization)
Analogy	Student in exactly one class	Student 70% Class A, 30% Class B

Chapter 16 Roadmap

Section	Topic
16.1	Applications of clustering in IR
16.2	Problem definition
16.3	Evaluating cluster quality
16.4	K-Means algorithm (hard clustering)

Key Points to Remember

- ✓ Clustering $\neq$ Classification: No labels, no teacher in clustering
- ✓ Distance measure = most critical design choice in clustering
- ✓ K-Means = hard + flat; EM = soft + flat
- ✓ Goal: maximize intra-cluster similarity, minimize inter-cluster similarity

Section 16.1: Clustering in Information Retrieval

The Cluster Hypothesis

Cluster Hypothesis: Documents in the same cluster behave similarly with respect to relevance to information needs.

If one document in a cluster is relevant to a query, it is very likely that other documents from the same cluster are also relevant — because clustering puts together documents that share many terms. This is essentially the *contiguity hypothesis* from Chapter 14.

Applications of Clustering in IR (Table 16.1)

Application	What is Clustered?	Benefit	Example
Search Result Clustering	Search results (returned docs)	Better presentation to user	Vivísimo engine
Scatter-Gather	Whole collection	Alternative UI — no typing needed	Figure 16.3
Collection Clustering	Whole collection	Exploratory browsing	Google News
Cluster-Based Retrieval	Whole collection	Better precision AND recall	Liu & Croft (2004)
Faster Search	Whole collection	Speed — fewer comparisons needed	Salton (1971)

Application 1: Search Result Clustering

Normal search returns a flat ranked list — confusing when a term is ambiguous. Example: 'jaguar' can mean the car brand, the animal, or Apple's OS X version. Clustering groups results by sense so users can directly click the relevant cluster, improving recall even if a sense is absent from top results.

Application 2: Scatter-Gather

Scatter = cluster whole collection into broad groups. **Gather** = user picks interesting groups, merges them, and scatters again. Repeat until desired cluster is found. No typing required — just clicking! *Analogy: zooming into a map from world → continent → country → city.*

Application 3: Collection Clustering (Static)

Pre-compute clusters of the entire collection (e.g., Google News). Similar stories are grouped together. Clusters recomputed frequently for news. News reading is browsing, not searching — clustering suits this well.

Application 4: Cluster-Based Retrieval (Improving Recall)

Query: 'car'. Relevant doc uses 'automobile' — standard search misses it! Clustering fix: find initial matching docs → identify their cluster → add other docs from same cluster. These may use synonyms → recall increases! Also used in language modeling: smooth document model with cluster model instead of whole collection.

Application 5: Faster Search (Efficiency)

Instead of comparing query to ALL documents: (1) Compare query to cluster centroids (much fewer). (2) Find closest cluster(s). (3) Only search within those clusters. (4) Rank and return. This is inexact but the expected quality loss is small while speed gain is huge.

Key Points to Remember

- ✓ Cluster Hypothesis = the theoretical justification for ALL clustering applications in IR
- ✓ Search result clustering helps with ambiguous queries (like 'jaguar')
- ✓ Scatter-Gather = iterative, user-driven browsing — no keywords needed
- ✓ Cluster-based retrieval finds docs with synonyms the query didn't use
- ✓ Clustering speeds up search by reducing the search space

Common Confusions

Confusion	Clarification
Does cluster-based retrieval always improve results?	Not always — cluster hypothesis can fail (see Exercise 16.1)
Is scatter-gather = hierarchical clustering?	No — scatter-gather is user-driven and interactive
Why is cluster-based search inexact?	You only search the nearest cluster — may miss relevant docs elsewhere

Exercises (16.1 & 16.2)

Exercise 16.1: Find a case where the cluster hypothesis fails.

If two documents share proper names like 'Clinton' and 'Sarkozy' — one may be about climate policy and the other about a football match. Sharing names ≠ sharing topics → cluster hypothesis breaks down.

Exercise 16.2: 1D example where cluster-based retrieval does worse than direct search.

A query point sitting exactly between two clusters — the nearest cluster might not contain the true nearest document. Direct nearest-neighbor search would find it; cluster-based retrieval misses it.

Section 16.2: Problem Statement

The Formal Goal of Hard Flat Clustering

Three Inputs Required:

Input	Symbol	Meaning
Set of documents	$D = \{d_1, d_2, \dots, d_N\}$	All N documents to be clustered
Number of clusters	K	How many groups we want
Objective function	f	A formula that measures clustering quality

The Assignment Function γ (gamma)

$\gamma: D \rightarrow \{1, 2, \dots, K\}$ is a function that assigns every document to a cluster number (1 through K). Every document gets exactly one label. The goal is to minimize (or maximize) the objective function. For K-Means: minimize average distance from each document to its cluster centroid.

Surjective Requirement

γ must be **surjective** — every cluster must have at least one document. No empty clusters allowed! Bad: Cluster 3 = $\{\}$ (empty). Good: every cluster has ≥ 1 document.

Similarity vs Distance

Measure	What it tells us	Better when...
Similarity (e.g., cosine)	How alike two documents are (0 to 1)	We want to maximize closeness
Distance (e.g., Euclidean)	How far apart two documents are	We want to minimize separation

Topic Similarity Examples

Topic	High-value dimensions (important words)
China	'Chinese', 'Beijing', 'Mao'
United Kingdom	'London', 'Britain', 'Queen'

Stop Words — Special Case

Usually stop words are ignored in IR. Exception: when clustering by **language**. In English, 'the' is very frequent and 'la' is rare. In French, 'la' is frequent and 'the' is rare. So stop words become useful signals for language-based clustering.

Terminology Clarification

Term	Meaning
Hard clustering	Membership values are 0 or 1 only
Soft clustering	Membership can be any non-negative value (fractions)
Partitional clustering	Each document belongs to exactly one cluster
Exhaustive clustering	Every document is assigned to some cluster (none left out)

Nonexhaustive clustering	Some documents may be assigned to no cluster
This book's definition	Exhaustive — every document must be assigned

Common Confusions

Confusion	Clarification
What exactly is a centroid?	The average vector of all documents in a cluster — the cluster's center point
Why specify K in advance?	Limitation of flat clustering; choosing K wisely is a key challenge
Is partitional = hard clustering?	Not exactly — partitional means one cluster per doc; hard means 0/1 membership

Section 16.2.1: Cardinality — The Number of Clusters (K)

How Do We Choose K?

Method	Description	Reliability
Experience/Domain knowledge	Expert guesses based on the problem	Subjective but practical
Application constraints	System/interface limits K (e.g., screen size)	External, hard constraint
Heuristic methods	Mathematical tricks to estimate K	Better, but not perfect
Incorporate K into objective function	Let the algorithm figure out K	Most principled approach

Clustering as a Search Problem

Since the goal is to optimize an objective function, clustering is essentially a **search problem**. We search through all possible assignments of N documents into K clusters looking for the best one.

Why Not Brute Force?

The number of possible clusterings grows **exponentially** with N and K. For N=100 documents and K=10 clusters, the number of possible partitions is incomprehensibly large. Even supercomputers can't handle this — brute force is completely infeasible.

The Practical Solution: Iterative Refinement

Step	Action
Step 1	Start with an INITIAL partitioning (some starting assignment)
Step 2	Iteratively REFINE — keep improving the assignment
Step 3	Stop when the objective function can't be improved further

Global vs Local Optimum

Iterative refinement is like climbing a hill in the dark — you can't see the whole landscape, so you keep stepping upward from your starting point. If you start near a *small hill*, you reach its top (local optimum) and think you're done — but the actual mountain (global optimum) is far away and you never found it.

Term	Meaning
Global Optimum	The absolute best clustering possible
Local Optimum	Looks best from nearby starting points, but not globally best

Key Points to Remember

- ✓ K must often be guessed or constrained externally — no universal formula
- ✓ Number of possible clusterings grows exponentially — brute force is out
- ✓ All practical flat clustering algorithms use iterative refinement
- ✓ Starting point quality directly impacts final clustering quality
- ✓ This is why K-Means is run multiple times with different starting points in practice

Section 16.3: Evaluation of Clustering

Internal vs External Criteria

Criterion	Based on	Problem
Internal	Data itself — intra-cluster similarity & inter-cluster dissimilarity	Good internal score $\neq$ good real-world usefulness
External	Compare to gold standard (human-labeled reference classes)	Requires expensive human annotation

Figure 16.4 — Running Example

17 documents, 3 clusters, 3 classes (x, o, ■):

Cluster	Documents	Majority Class	Majority Count
Cluster 1 (6 docs)	x x x x x o	x	5
Cluster 2 (6 docs)	o o o o x ■	o	4
Cluster 3 (5 docs)	■ ■ ■ x o	■	3

Measure 1: Purity

For each cluster, find the most frequent class (majority class). Count docs belonging to that class. Divide total by N.

$$\text{Formula: } \text{purity}(\Omega, C) = (1/N) \times \sum_k [\max_j |\omega_k \cap c_j|]$$

Step-by-Step Calculation:

Step	Action	Value
1	Majority count: Cluster 1 = 5 (x), Cluster 2 = 4 (o), Cluster 3 = 3 (■)	5, 4, 3
2	Sum of majority counts	$5 + 4 + 3 = 12$
3	Divide by $N=17$	$12/17 \approx 0.71$

Range: 0 (worst) to 1 (perfect). **Critical weakness:** Purity = 1 if each document gets its own cluster ($K=N$)! Cannot detect this trivially perfect but useless clustering.

Measure 2: Normalized Mutual Information (NMI)

Why needed: To fix purity's weakness — NMI penalizes clusterings with too many clusters.

$$\text{NMI}(\Omega, C) = I(\Omega; C) / [(H(\Omega) + H(C)) / 2]$$

$$\text{Mutual Information: } I(\Omega; C) = \sum_k \sum_j (|\omega_k \cap c_j|/N) \times \log(N \times |\omega_k \cap c_j| / (|\omega_k| \times |c_j|))$$

$$\text{Entropy: } H(\Omega) = -\sum_k (|\omega_k|/N) \times \log(|\omega_k|/N)$$

Variable Dictionary:

Symbol	Meaning
$I(\Omega; C)$	Mutual information — how much knowing a cluster tells us about the class

$H(\Omega)$	Entropy of cluster assignments — uncertainty/spread of clusters
$H(C)$	Entropy of class assignments — uncertainty/spread of classes
$[H(\Omega)+H(C)]/2$	Normalization term — ensures NMI stays between 0 and 1

When $K=N$: $H(\Omega)=\log N$ (maximum), making denominator large $\rightarrow$ NMI stays low $\rightarrow$ large K is penalized! NMI can be used to compare clusterings with different values of K (purity cannot).

Measure 3: Rand Index (RI)

Treats clustering as pairwise decisions over all $N(N-1)/2$ pairs of documents. For each pair: should they be in the same cluster or different clusters?

Decision	Meaning	Good or Bad?
TP (True Positive)	Two similar docs $\rightarrow$ same cluster	Correct
TN (True Negative)	Two dissimilar docs $\rightarrow$ different clusters	Correct
FP (False Positive)	Two dissimilar docs $\rightarrow$ same cluster	Wrong
FN (False Negative)	Two similar docs $\rightarrow$ different clusters	Wrong

$$RI = (TP + TN) / (TP + FP + FN + TN)$$

Step-by-Step Calculation for Figure 16.4:

Step	Calculation	Result
TP+FP (same cluster pairs)	$C(6,2) + C(6,2) + C(5,2) = 15+15+10$	40
TP (correct same-cluster)	$C(5,2)+C(4,2)+C(3,2)+C(2,2) = 10+6+3+1$	20
FP	$40 - 20$	20
FN (from contingency table)	Given	24
TN	$136 - 20 - 20 - 24$	72
RI	$(20+72)/(20+20+24+72) = 92/136$	≈ 0.68

Measure 4: F-Measure

Like RI but lets you penalize FN more than FP using parameter β . $\beta > 1$ means recall matters more (penalize separating similar documents more).

$$P = TP/(TP+FP) \quad R = TP/(TP+FN) \quad F\beta = (\beta^2+1) \times P \times R / (\beta^2 \times P + R)$$

Metric	Calculation	Value
Precision P	$20/40$	0.5
Recall R	$20/44$	≈ 0.455
F1 ($\beta=1$)	$2 \times 0.5 \times 0.455 / (0.5 + 0.455)$	≈ 0.48
F5 ($\beta=5$)	Heavily favors recall	≈ 0.456

All Four Measures — Table 16.2

Measure	Lower Bound	Maximum	Figure 16.4 Value	Handles large K?
Purity	0.0	1.0	0.71	No — inflated by large K
NMI	0.0	1.0	0.36	Yes — entropy normalization
Rand Index	0.0	1.0	0.68	Yes
F5	0.0	1.0	0.46	Yes

Common Confusions

Confusion	Clarification
Why is NMI better than purity?	NMI penalizes large K through entropy normalization; purity doesn't
What does mutual information measure?	How much knowing the cluster reduces uncertainty about the class
Why use pairs of documents in RI?	Reframes clustering as binary decisions — easier to apply precision/recall
Can NMI be greater than 1?	No — normalization guarantees it's always between 0 and 1

Exercise 16.3

Q: Replace every point d in Figure 16.4 with two identical copies. (i) Harder, equally, or easier to cluster 34 points vs 17? (ii) Compute purity, NMI, RI, F5 for 34 points. (iii) Which measures best compare quality?

A: (i) Equally difficult — the cluster structure is identical, just doubled. (ii) Purity stays 0.71 (ratio unchanged); NMI stays same (ratios unchanged); RI stays ~0.68; F5 stays ~0.456. All measures are ratio-based so doubling identical points doesn't change them. (iii) NMI is best — it's normalized and handles different K values properly.

Section 16.4.1: Cluster Cardinality in K-Means

What is RSS?

RSS (Residual Sum of Squares) = total squared distance of every document from its cluster's centroid. Low RSS = documents close to centroids = tight, good clusters. High RSS = loose, bad clusters. K-Means tries to minimize RSS — this is its core objective function.

$$\text{RSS} = \sum_k \sum_{(d \in \omega_k)} |d - \mu_k|^2$$

where ω_k = cluster k , μ_k = centroid of cluster k , $|d - \mu_k|^2$ = squared distance from doc to centroid.

Three Approaches to Choosing K

Approach 1: Naive — Minimize RSS Directly (FAILS)

$\text{RSS}_{\min}(K)$ is **monotonically decreasing** in K . More clusters always means lower RSS. At $K=N$, every doc is its own centroid $\rightarrow \text{RSS}=0$. Naive minimization always gives $K=N$ — completely useless!

K	RSS behaviour	Problem?
K=1	All docs in one cluster — very high RSS	No clustering
K=2	Two clusters — lower RSS	OK
K=N	Each doc in own cluster — RSS=0	Useless! Trivial solution

Approach 2: Knee/Elbow Heuristic (PRACTICAL)

Step	Action
Step 1	For each K: run K-Means $i=10$ times with different initializations
Step 2	Record the MINIMUM RSS found across i runs $\rightarrow \text{RSS}_{\min}(K)$
Step 3	Plot $\text{RSS}_{\min}(K)$ vs K
Step 4	Find the 'KNEE' — where the curve flattens out (improvements shrink)

Figure 16.8 shows two knees for 1,203 Reuters-RCV1 documents: one at $K=4$, another at $K=9$. The documents came from 4 categories (China, Germany, Russia, Sports) so $K=4$ is the natural choice. Caveat: there is seldom a single best K — multiple knees are common.

Why run $i=10$ times? K-Means is sensitive to initialization — different starts give different RSS. Taking the minimum across 10 runs gives a reliable $\text{RSS}_{\min}(K)$ estimate.

Approach 3: Penalized Objective Function

$$K = \text{argmin}_K [\text{RSS}_{\min}(K) + \lambda K]$$

Symbol	Meaning
$\text{RSS}_{\min}(K)$	Best RSS achievable with K clusters (distortion)
λK	Penalty for model complexity ($\lambda \times$ number of clusters)
λ (lambda)	Weighting factor — controls how much to penalize extra clusters

Value of λ	Effect
$\lambda = 0$	No penalty $\rightarrow K=N$ wins (useless)
λ very large	Heavy penalty $\rightarrow K=1$ wins (also useless)
λ just right	Balanced $\rightarrow$ meaningful K found

AIC — Theoretical Justification

$$\text{General AIC: } K = \operatorname{argmin}_K [-2L(K) + 2q(K)]$$

$$\text{AIC for K-Means: } K = \operatorname{argmin}_K [\text{RSSmin}(K) + 2MK]$$

Term	Meaning
$-2L(K)$	Negative log-likelihood = distortion (how poorly model fits data)
$2q(K) = 2MK$	$2 \times$ number of parameters = model complexity (K centroids $\times$ M dimensions each)
$\lambda = 2M$	AIC is Equation 16.11 with this specific penalty weight

Why AIC Fails for Text Clustering:

For text, $M \approx 50,000$ (very high-dimensional). AIC penalty = $2MK = 100,000K$. But $\text{RSSmin}(1) < 5,000$. The penalty dominates completely $\rightarrow$ AIC always picks $K=1$. AIC assumes low-dimensional data — impractical for raw text vectors.

Summary: Three Approaches to Choosing K

Approach	Method	Verdict
Naive	Minimize RSS directly	FAILS — always gives $K=N$
Knee heuristic	Find elbow in RSS curve	Practical, but subjective
Penalized objective	$\text{RSS} + \lambda K$	Principled, but λ needs tuning
AIC	$\text{RSS} + 2MK$	Theoretically nice, fails for high-dim text

Key Points to Remember

- ✓ RSS always decreases as K increases — naive minimization gives $K=N$ (useless)
- ✓ Run K-Means i times per K to get reliable $\text{RSSmin}(K)$ estimates
- ✓ The knee/elbow in RSS curve gives candidate values of K
- ✓ AIC = distortion + $2 \times$ model complexity; $\lambda = 2M$ for K-Means
- ✓ For high-dimensional text data, M is huge $\rightarrow$ AIC penalty dominates $\rightarrow$ AIC breaks down

Common Confusions

Confusion	Clarification
Why does RSS always decrease with K ?	More clusters = each doc closer to centroid. At $K=N$, $\text{RSS}=0$ trivially
Is the knee method reliable?	It's a heuristic — works well in practice but requires judgment
Is AIC always wrong for IR?	For low-dimensional representations it can work; fails for raw high-dim text
Difference between λ and $2M$?	λ is tunable; $2M$ is what AIC derives from theory for K-Means

Exercises (16.4 & 16.5)

Exercise 16.4: Why do documents not using the same term for 'car' likely end up in the same cluster?

K-Means groups documents by overall vector similarity. Documents about cars — even if they use 'automobile', 'vehicle', 'sedan' instead of 'car' — share many other related terms (engine, speed, road, driver, etc.). Their overall vectors are similar → they cluster together despite term variation.

Exercise 16.5: Two termination conditions: (i) assignment doesn't change, (ii) centroid doesn't change — are they equivalent?

Yes, they are equivalent. If assignments don't change, the set of documents in each cluster doesn't change, so centroids (computed as averages) also don't change. Conversely, if centroids don't change, every document's nearest centroid stays the same, so assignments don't change either.

Quick Reference: All Formulas

Formula	Name	Purpose
$\text{purity}(\Omega, C) = (1/N) \sum_k \max_j \omega_k \cap c_j $	Purity	Simple cluster accuracy
$\text{NMI}(\Omega, C) = I(\Omega; C) / [(H(\Omega) + H(C))/2]$	NMI	Info-theoretic, handles large K
$I(\Omega; C) = \sum_k \sum_j (\omega_k \cap c_j /N) \log(N \omega_k \cap c_j / \omega_k c_j)$	Mutual Information	Info gain from clusters about classes
$H(\Omega) = -\sum_k (\omega_k /N) \log(\omega_k /N)$	Entropy	Uncertainty of cluster assignments
$\text{RI} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{FN} + \text{TN})$	Rand Index	Accuracy over all document pairs
$F_\beta = (\beta^2 + 1) \text{PR} / (\beta^2 \text{P} + \text{R})$	F-Measure	Weighted precision-recall balance
$\text{RSS} = \sum_k \sum_{d \in \omega_k} d - \mu_k ^2$	RSS	K-Means objective — total distortion
$K = \text{argmin}_K [\text{RSSmin}(K) + \lambda K]$	Penalized K selection	Balance distortion vs complexity
AIC: $K = \text{argmin}_K [\text{RSSmin}(K) + 2MK]$	AIC for K-Means	Theoretical K selection ($\lambda=2M$)